USN | | | | | | | | | | [23TCSE301]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER -2024

OPERATING SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5x2=10**		
1.	**a.**	Explain the primary function of system calls in an operating system?	L2	CO1	**02**
	b.	Outline the critical-section problem and deadlock	L2	CO2	**02**
	c.	Illustrate page fault	L2	CO3	**02**
	d.	What are I/O hardware components and their needs?	L2	CO4	**02**
	e.	What is an Linux system call?	L2	CO5	**02**
		PART-B	**6X5=30**		
2.	**a.**	Compare the evolution of operating systems from early single-tasking systems to modern multitasking systems.	L3	CO1	**05**
		OR			
	b.	Explain interrupt, and how does it affect the CPU's normal operation?	L3	CO1	**05**
3.	**a.**	Demonstrate the concept of multithreading and the different models of multithreading.	L3	CO2	**05**
		OR			
	b.	Outline deadlock prevention strategies and how they address the four conditions necessary for deadlock.	L3	CO2	**05**
4.	**a.**	Define thrashing, and How can it be mitigated?	L2	CO3	**05**
		OR			
	b.	Explain page replacement policy	L2	CO3	**05**
5.	**a.**	What is swap space management, and interpret the benefits and drawbacks of using swap space in operating systems?	L3	CO4	**05**
		OR			
	b.	Sketch the key techniques used for file system recovery.	L3	CO4	**5**
6.	**a.**	Describe the core functions of the Core OS layer in iOS	L2	CO5	**05**
		OR			
	b.	Explain the common inter-process communication (IPC) mechanisms in Linux, and how do they work?	L2	CO5	**05**
7.	**a.**	Explain the terms Mutex locks, Semaphores, Critical regions, Monitors, Multiple- processor scheduling	L2	CO3	**05**
		OR			
	b.	Explain the terms system boot, Threading, Kernel I/O subsystem,	L2	CO2	**05**

		Demand Paging, Recovery			
		PART-C	**6X10=60**		
8.	**a.**	Describe the memory hierarchy in modern computer systems. How does each level of the hierarchy (registers, cache, RAM, and storage) contribute to system performance.	L3	CO1	**10**
		OR			
	b.	Explain Direct Memory Access process and its advantages over traditional I/O operations.	L4	CO1	**10**
9.	**a.**	Outline the role of process scheduling in an operating system. and how do they impact system performance?	L3	CO2	**10**
		OR			
	b.	Identify the issues encountered in multithreaded applications, such as race conditions, deadlock, and thread starvation. Explain strategies to address these issues, including the use of synchronization mechanisms and design practices.	L2	CO2	**10**
10.	**a.**	Explain the concept of virtual memory; how the challenges associated with managing virtual memory in modern operating systems	L4	CO3	**10**
		OR			
	b.	Explain the concept of segmentation and paging. How does segmentation differ from paging	L4	CO3	**10**
11.	**a.**	Explain the disk structure and how different disk scheduling algorithms improve system performance.	L4	CO4	**10**
		OR			
	b.	Discuss free space management techniques in file systems and their effectiveness.	L2	CO4	**10**
12.	**a.**	Explain the role and components of the Android SDK framework in mobile application development.	L4	CO5	**10**
		OR			
	b.	Describe the Linux file system structure- Root Directory, File System Hierarchy, Inodes and Data Blocks, File types	L3	CO5	**10**
13.	**a.**	Explain the basic elements of a computer system and their roles. How do these elements interact during instruction execution?	L4	CO1	**10**
		OR			
	b.	Discuss the mass Storage Structure and Application I/O interface in detail	L4	CO5	**10**